CS585 Project

When Should We Trust Medical Image Models?

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Research Question Recap

Medical image models often hit high AUC on benchmarks

But: does their **CONFIDENCE** transfer across hospitals?

Two chest pathologies:

- Pneumothorax
- Pleural Effusion(PE)

Two CNN Models

Calibration method: Temperature Scaling

Pipeline We Built

Chest X-ray datasets :

CheXpert(Stanford) :
191K training samples, U-Ones policy

MIMIC CXR: (Boston)
5,159 test dicoms → 1,835 labeled frontal cross-domain test samples

Models:

ResNet-50:
ImageNet pretrained

DenseNet-121:
ImageNet pretrained

Evaluation:

Calibration metrics + Bootstrap confidence intervals(CI)

Cross-domain:
5 settings × 2 models

Grad-CAM:
mechanism analysis

Baseline Performance (CheXpert)

Model	Pneumothorax AUC	Pleural Effusion AUC	Mean AUC
ResNet-50	0.9136	0.9322	0.9229
DenseNet-121	0.9348	0.9258	0.9303

- Adam lr=1e-4, 10 epochs, BCE With Logits, ImageNet-pretrained
- DenseNet slightly better overall, but strengths differ per pathology

Finding 1: Cross-Domain Degradation — AND Rank Reversal

Pathology	Model	In-domain	Cross-domain	Drop
Pneumothorax	ResNet-50	0.914	**0.836**	-0.078
Pneumothorax	DenseNet-121	0.935	0.810	**-0.125**
Pleural Effusion	ResNet-50	0.932	0.855	-0.077
Pleural Effusion	DenseNet-121	0.926	0.853	-0.073

Key insight: "In-domain best ≠ Cross-domain best"

Finding 2: TS Improves ECE but Hurts Safety Metric

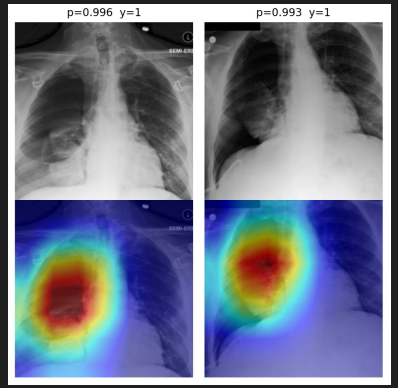
Uncalibrated → Temperature-Scaled

ResNet Pneumothorax:
ECE: 0.022 → 0.028 (slightly worse)
Conf-on-incorrect: 0.833 → 0.864 ⚠️ WORSE

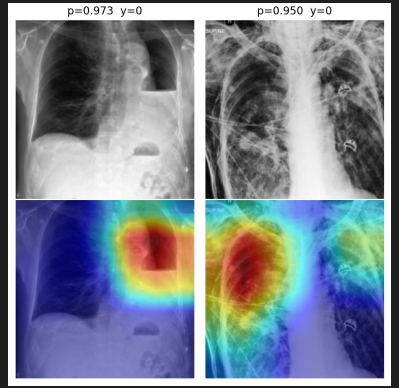
All 4 settings (2 models × 2 pathologies):
Conf-on-incorrect INCREASES in 4/4 (CI non-overlap)

Finding 3: Grad-CAM Reveals Mechanism Behind Rank Reversal

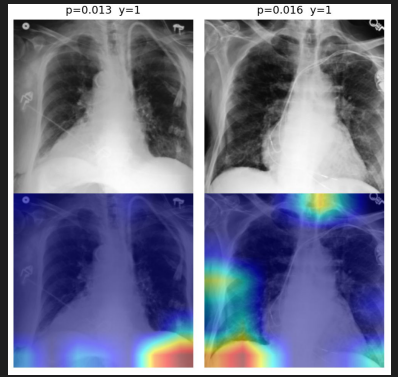
DenseNet
TP:
spatial prior



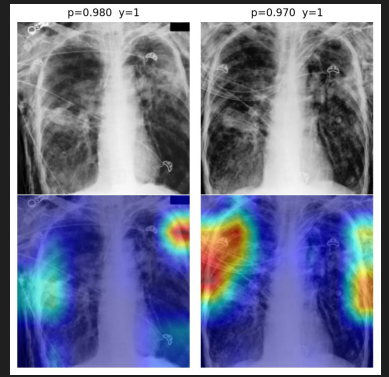
DenseNet
FP:
spatial prior



DenseNet
FN:
Attention on
image edges



ResNet TP:
Distributed,
no strong
bias



Additional Findings

- Pathology-specific calibration shift: ResNet Pneumothorax ECE improves cross-domain (0.080→0.022); Pleural Effusion worsens (0.100→0.181). Not uniform.
- Counter-intuitive subset behavior: shared_only subset has WORSE ECE than extra_pathology subset. Samples with co-occurring pathologies are actually closer to CheXpert training distribution.
- DenseNet Pneumothorax recall deficit: only 18/108 true positives reach prob≥0.9; 35/108 get prob≤0.1 (confident misses).

Conclusion

- **In-domain AUC on CheXpert is not a good proxy for MIMIC behavior:** we see AUC drops under shift, and the stronger in-domain model is not always stronger out-of-domain (especially clear for Pneumothorax in our “all MIMIC” results).
- **Calibration and discrimination split:** Pleural Effusion shows much higher ECE on MIMIC than Pneumothorax in our numbers—reliability is pathology-dependent, not a single “model is calibrated” label.
- **Temperature scaling fit on CheXpert validation is not a free fix:** ECE is mixed after TS, but confidence on incorrect predictions moves in the wrong direction in all model–disease point estimates we report.
- **Bottom line:** Deploying clinical AI should use cross-site testing plus safety-oriented metrics, not benchmark AUC alone.

Future Steps

- **Pleural Effusion / Pneumothorax–specific validation:** separate calibration and error analysis per pathology (e.g. PE: prevalence, ECE; Pneumothorax: rare positives, unstable AUC CIs, high-confidence failures).
- **Target-domain calibration:** fit temperature (or vector/Platt) scaling on a MIMIC (or new hospital) validation split, and compare to source-only TS.
- **Cohort / label design:** expand shared-only vs co-occurring disease subsets; pre-register how subsets affect AUC vs ECE so subset stories stay interpretable.
- **Modeling:** reduce spurious cues suggested by Grad-CAM (e.g. augmentations, regularization, ensembles) and re-check ranking and confidence on errors under shift.
- **Translation:** small prospective multi-center or reader study with pre-set operating points from reliability metrics, not AUC only.